SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO5_AT3 — INTERACTIVE DASHBOARD PROJECT

Course Code:	DSA06	**Course Title:**	Data Handling and Visualization
CO Assessed:	CO5 – Create (BL6)	**Assessment:**	Assessment Tool 3 – Interactive Dashboard Project
Weightage:	20% of CIA	**Total Marks:**	25 (Dataset: 7, Interactivity: 9, Effective Visualization: 9)
CO5 Statement:	*Create an original, interactive dashboard that selects and prepares an appropriate dataset, implements meaningful software interactivity, and applies effective visualization design principles.(BL6)*		
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Section / Batch:		**Date:**	29.08.2026

Code of Conduct

I Priyanka.G/ 192524124 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ________________________

Instructions

- This is an individual project. Each student designs, builds, and submits one original interactive dashboard. Total: 25 Marks.
- Electronic devices and software tools are expected and permitted — use any dashboarding tool (e.g., Tableau, Power BI, Plotly Dash, Streamlit, or similar) of your choice.
- Late submissions and academic integrity are governed by the standard course policy; all code/data sources must be cited.
- Submission window: as announced by the instructor (project, not a timed test).
- Choose and clearly justify your own dataset — it must be real or realistic, relevant to a stated purpose, and appropriately cleaned/prepared before use.
- The dashboard must include genuine software interactivity (e.g., filters, zoom/pan, hover tooltips, drill-down, or linked views) — a static image or PDF does not satisfy this requirement.
- Apply effective visualization principles covered in this course (proportional ink, appropriate axis/color scale choice, avoiding overplotting, accessibility) throughout the dashboard.
- Submit both the working dashboard (or a recorded walkthrough, if live hosting is not possible) and a short written justification of your dataset and design choices.

Annexure A – INTERACTIVE DASHBOARD PROJECT

Faculty In-charge	Course Coordinator	Head of Department
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________

Project Title: COVID-19 Pandemic Trend Analysis

1. Project Overview

The **COVID-19 Pandemic Trend Analysis** project is an interactive web-based dashboard developed using **Python and Streamlit** for exploring and analyzing COVID-19 pandemic data.

The main purpose of the project is to transform pandemic-related datasets into an interactive analytical environment where users can examine global trends, compare countries, analyze India's pandemic situation, investigate peak periods, study vaccination information, and derive meaningful insights from the available data.

Instead of presenting the information through static charts or tables, the dashboard provides multiple analytical views that allow users to navigate between different aspects of the pandemic.

The application follows a structured dashboard design with separate analytical sections including:

- Overview
- Global Analysis
- Country Analysis
- India Analysis
- Pandemic Trends
- Peak Analysis
- Vaccination Analysis
- Country Comparison
- Country Rankings
- Data Explorer
- Insights
- About

The dashboard also includes an authentication interface and a protected workspace structure.

2. Problem Statement

The COVID-19 pandemic generated large volumes of data containing information about cases, deaths, trends, peaks, countries and vaccination coverage.

However, raw datasets are difficult to interpret directly because they contain large numbers of records and multiple variables.

The objective of this project is therefore to develop an interactive dashboard that:

- Organizes COVID-19 data into meaningful analytical categories.
- Allows users to explore pandemic trends interactively.
- Enables country-level and global comparisons.

- Provides focused analysis of India's pandemic situation.
- Identifies important pandemic peaks and trends.
- Presents vaccination-related information.
- Communicates findings through effective visualizations.
- Converts raw pandemic data into understandable data-driven insights.

3. Objectives

The major objectives of the project are:

3.1 Data Analysis

To analyze COVID-19 pandemic data and identify meaningful trends, patterns and differences between countries and time periods.

3.2 Interactive Exploration

To provide users with an interactive dashboard where they can navigate between different analytical views rather than relying on a static report.

3.3 Comparative Analysis

To enable comparison between countries using metrics such as reported cases, deaths and other available pandemic indicators.

3.4 Peak Analysis

To identify and analyze periods in which significant increases in reported COVID-19 activity occurred.

3.5 Vaccination Analysis

To provide a dedicated analytical view of vaccination-related information available in the dataset.

3.6 Data Visualization

To communicate numerical information using charts, metrics, tables and structured dashboard components.

3.7 Data-Driven Insights

To convert the analyzed data into understandable observations that can assist users in interpreting pandemic patterns.

4. Dataset Selection, Quality and Preparation

4.1 Dataset Selection

The project uses COVID-19 pandemic surveillance datasets stored as project CSV files.

The selected data is relevant to the project because it contains information required to study the progression and impact of the COVID-19 pandemic across different geographic regions and time periods.

The datasets are organized into analytical categories used by the dashboard, including:

- Country-level pandemic data
- Global pandemic data
- Peak analysis data
- Vaccination data

The dataset was selected because COVID-19 provides a suitable real-world context for demonstrating data handling, exploratory analysis, comparison, trend identification and visualization.

4.2 Relevance of the Dataset

The dataset directly supports the stated purpose of the dashboard.

It allows the application to answer questions such as:

- How did COVID-19 trends change over time?
- Which countries experienced higher reported case counts?
- How did different countries compare?
- What were the major pandemic peaks?
- How did India compare with global trends?
- What information is available regarding vaccination coverage?
- What patterns can be identified from the reported surveillance data?

Therefore, the dataset is neither a toy dataset nor an arbitrary collection of values. It represents a real-world public-health data analysis problem.

4.3 Data Preparation

Before visualization, the project processes the CSV datasets through the application's data-loading and analysis components.

The preparation process includes:

- Loading the required CSV files.
- Organizing the datasets according to their analytical purpose.

- Converting the data into pandas DataFrames.
- Preparing variables required for analysis.
- Separating global, country, peak and vaccination information.
- Preparing data for chart generation and comparisons.
- Handling the datasets through reusable data-loading functions.

The dashboard uses a centralized data-loading approach so that the visualization pages can work with the prepared datasets consistently.

4.4 Dataset Limitations

The dashboard is based on **reported surveillance data**, and therefore the results should be interpreted as reported observations rather than a complete representation of every COVID-19 infection.

Potential limitations include:

- Differences in testing practices between countries.
- Differences in reporting standards.
- Missing or incomplete observations in some periods.
- Differences in vaccination reporting methodologies.
- Changes in data collection practices over time.
- Reported values may not represent the actual total number of infections.

These limitations are important when interpreting country-level comparisons and pandemic trends.

5. Dashboard Architecture

The application was developed using a modular structure.

The major components include:

```
COVID-19-Pandemic-Trend-Analysis
│
├── auth/
│   └── Authentication module
│
├── dashboard/
│   ├── app.py
│   ├── components/
│   │   └── styles.py
│   ├── utils/
```

```
│   │   └── data_loader.py
│   └── views/
│       └── Dashboard analysis pages
│
├── database/
│   └── Database and activity management
│
├── data/
│   └── COVID-19 CSV datasets
│
└── requirements.txt
```

This modular organization separates:

- Authentication
- Data loading
- Database operations
- Styling
- Dashboard views
- Application control

This makes the application easier to maintain and extend.

6. Software and Technologies Used

Technology	**Purpose**
Python	Core programming language
Streamlit	Interactive web dashboard framework
Pandas	Data processing and analysis
Plotly / visualization libraries used by project	Interactive data visualization
SQLite	Local authentication/session-related database functionality
HTML/CSS	Dashboard interface customization
Git	Version control
GitHub	Source-code repository and project deployment

7. Interactivity of the Dashboard

Interactivity is one of the major components of the project because the assessment specifically requires a functioning interactive dashboard rather than a static report.

The application provides multiple interactive analytical sections and controls.

7.1 Dashboard Navigation

The dashboard contains a navigation system that allows users to move between different analytical pages.

The available pages include:

1. Overview
2. Global Analysis
3. Country Analysis
4. India Analysis
5. Pandemic Trends
6. Peak Analysis
7. Vaccination Analysis
8. Country Comparison
9. Country Rankings
10. Data Explorer
11. Insights
12. About

This allows users to choose the type of analysis they want to perform.

7.2 Country Analysis

The Country Analysis section allows users to focus on individual countries and examine their pandemic-related information.This is useful because global aggregates can hide significant differences between countries.The user can therefore move from a global perspective to a more detailed country-level perspective.

7.3 India Analysis

A dedicated **India Analysis** section is provided to examine India's pandemic situation separately.

This provides a focused analytical view rather than requiring users to identify India manually from a global dataset.

The section can be used to examine India's reported trends alongside relevant global and peak information.

7.4 Country Comparison

The Country Comparison page provides an interactive method for comparing countries.This allows users to investigate differences between selected countries rather than examining each country independently.

Country comparison is particularly useful for identifying:

- Differences in reported cases
- Differences in reported deaths
- Relative pandemic impact
- Differences in trends

7.5 Country Rankings

The Country Rankings section organizes countries according to available pandemic metrics.This provides a quick way for users to identify countries with comparatively higher or lower reported values.Ranking is more efficient for analytical exploration than manually scanning a large dataset.

7.6 Data Explorer

The Data Explorer provides a more direct method of examining the underlying data.It supports users who want to move beyond summarized charts and inspect the available information in greater detail.This feature improves transparency because the dashboard does not rely only on pre-selected visual summaries.

7.7 Peak Analysis

The Peak Analysis page focuses specifically on major pandemic peaks.This allows users to examine periods of unusually high reported activity and understand how pandemic waves differed across locations.

7.8 Vaccination Analysis

The Vaccination Analysis section provides a dedicated view for exploring vaccination-related information.Separating vaccination analysis from infection trends makes it easier to study vaccination as a distinct pandemic-response variable.

8. Effective Visualization Design

The dashboard follows a medical-technology visual design intended to provide a professional and readable analytical environment.

The visualization design focuses on:

- Clear hierarchy
- Appropriate contrast
- Consistent styling
- Readable labels

- Meaningful analytical grouping
- Avoidance of unnecessary visual decoration
- Consistent dashboard structure

8.1 Appropriate Chart Selection

Different visual forms are used according to the type of information being communicated.

8.2 Trend Analysis

Line-based visualizations are appropriate for pandemic trends because time is an ordered variable.

They allow users to observe:

Increases

Decreases

Peaks

Waves

Changes over time

8.3 Country Comparison

Bar-based comparisons are suitable when the purpose is to compare numerical values between countries.

8.4 Rankings

Ranked visualizations make it easier to identify countries with relatively high or low values.

8.5 Tabular Data

Tables are useful where exact numerical values are more important than visual patterns.

9. Visualization Principles Applied

9.1 Proportional Representation

Visual elements are designed so that the displayed size and position of numerical information correspond meaningfully to the underlying values.

9.2 Appropriate Axes

Time-series visualizations use time as the ordered horizontal dimension, allowing the progression of pandemic activity to be interpreted naturally.

Axes and labels are used to provide context to numerical values.

9.3 Consistent Color Usage

The dashboard follows a consistent medical-tech color palette.

The interface uses a dark blue/teal visual identity with light analytical surfaces.

This provides:

- Strong visual hierarchy
- Consistent branding
- Good separation between dashboard sections
- Professional appearance

Colors are primarily used for emphasis and organization rather than unnecessary decoration.

9.4 Readability

The dashboard uses:

- Clear headings
- Consistent font styling
- Appropriate spacing
- Structured cards
- Readable labels
- Contrasting backgrounds

This helps users understand the information without being overwhelmed by visual clutter.

9.5 Avoiding Overplotting

The dashboard separates different analytical tasks into individual pages rather than attempting to display every variable on one chart.

For example:

- Global trends are separated from country analysis.
- Peak analysis is provided separately.
- Vaccination analysis has its own section.
- Country comparisons are separated from rankings.

This reduces visual complexity.

10. Dashboard User Flow

The intended user flow is:

Login / Account Creation
↓
Dashboard
↓
Overview
↓
↓ ↓ ↓
Global Analysis | Country Analysis | India Analysis
↓ ↓ ↓
Trends | Comparison | Peaks
↓
Vaccination
↓
Rankings
↓
Data Explorer
↓
Insights

This structure allows the user to move from general information to detailed analysis.

11. Authentication and Security

The application includes an authentication interface before access to the dashboard.

The authentication system provides:

- Sign-in functionality
- Account creation
- Username and password fields
- Password confirmation during registration
- Authentication state management
- Sign-out functionality
- SQLite-based account/session-related functionality

The project also uses password hashing through the authentication module rather than storing passwords directly as plain text.

The authentication system is primarily intended to demonstrate protected dashboard access within the academic project.

12. User Interface Design

The interface follows a medical analytics theme.

The authentication landing page contains:

- Academic analytics branding
- COVID-19 project title
- Description of the dashboard
- Global trends feature
- Country analysis feature
- Data insights feature
- Sign-in interface
- Account creation interface

The dashboard interface uses a structured sidebar containing:

- Dashboard branding
- Current user information
- Navigation
- Sign-out functionality
- Workspace status

This provides a consistent experience throughout the application.

13. Key Features of the Final Dashboard

The final application provides the following features:

Feature	Purpose
Authentication	Controls access to the dashboard
Overview	Provides a summary of pandemic information
Global Analysis	Examines global pandemic trends
Country Analysis	Provides country-specific analysis
India Analysis	Focuses on India's pandemic trends
Pandemic Trends	Examines changes over time
Peak Analysis	Identifies important pandemic peaks
Vaccination Analysis	Examines vaccination information
Country Comparison	Compares countries
Country Rankings	Ranks countries using available metrics
Data Explorer	Allows detailed data inspection
Insights	Presents analytical observations
About	Provides project information

15. Data-Driven Insights

The dashboard is designed not only to display charts but also to support interpretation of the data. The analysis can be used to identify observations such as:

Global Trends

> COVID-19 activity changes substantially over time, with periods of increasing and decreasing reported activity.

Country Differences

> Countries can show substantial differences in reported pandemic indicators due to population size, reporting practices, testing availability and pandemic progression.

Pandemic Peaks

> Major increases in reported activity can be identified as peaks, allowing different pandemic waves to be examined.

India

India can be analyzed separately to understand its pandemic progression and compare its patterns with global trends.

Vaccination

Vaccination data provides an additional dimension for understanding the pandemic response and comparing reported vaccination information between locations.

These observations demonstrate how the dashboard can be used as an analytical tool rather than simply as a data-display interface.

Conclusion

The **COVID-19 Pandemic Trend Analysis** project demonstrates the complete process of converting real-world pandemic data into an interactive analytical dashboard.The project combines data preparation, exploratory analysis, interactive software functionality and visualization design within a single Streamlit application.The dashboard allows users to move from global-level analysis to country-specific investigation, examine India's pandemic trends, identify peaks, explore vaccination information, compare countries, inspect data and interpret analytical insights.The project also demonstrates the application of effective visualization principles through appropriate chart selection, structured layouts, readable presentation and consistent visual design.Overall, the project fulfills the objectives of **CO5 — Create (BL6)** by creating an original interactive dashboard that selects and prepares an appropriate dataset, implements meaningful software interactivity and applies effective visualization design principles.

Future Enhancements

The following improvements could be considered in future versions:

1. Integration with a live COVID-19 data API.
2. Automated dataset updates.
3. Additional demographic analysis.
4. More advanced statistical analysis.
5. Predictive modelling of pandemic trends.
6. Additional vaccination effectiveness analysis.
7. Cloud-hosted database for persistent user accounts.
8. Export of filtered analytical results.
9. Downloadable charts and reports.
10. Deployment on a public cloud platform for wider access.

References

Dataset Source:
https://www.kaggle.com/datasets/caesarmario/our-world-in-data-covid19-dataset?select=owid-covid-data.csv

Software Documentation:
https://covid-19-pandemic-trend-analysis-qtsmwrvjvdmeebfmgn5mpt.streamlit.app/

Appendix

Appendix A: Dashboard

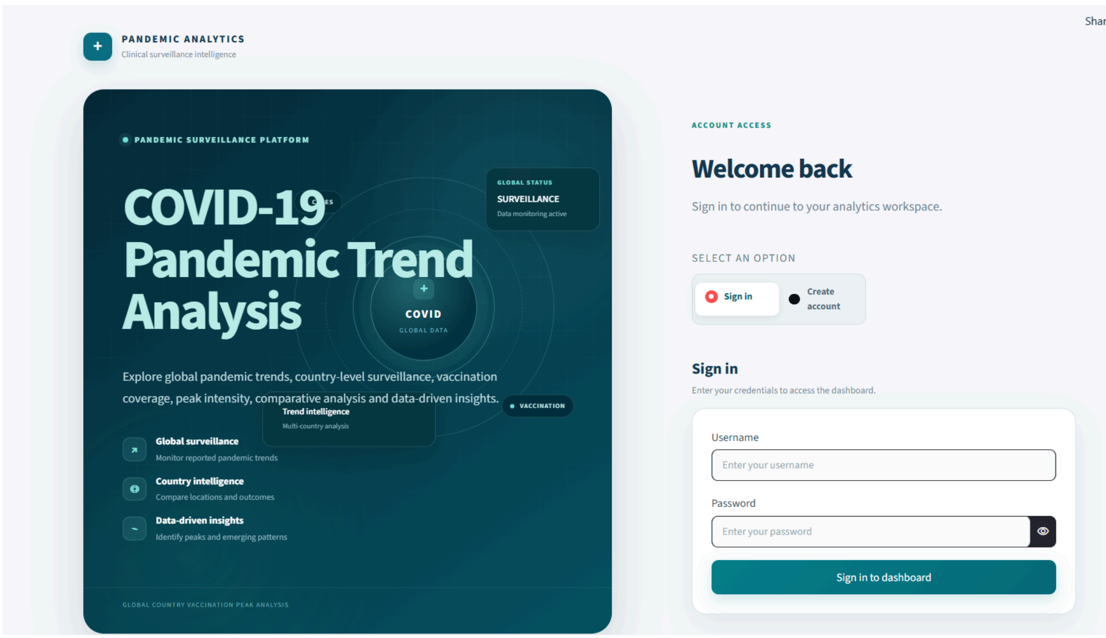

Figure: Covid-19 Analysis Dashboard

Appendix B: Source code : app.py

```
"""COVID-19 Pandemic Trend Analysis - authenticated Streamlit application."""

from __future__ import annotations

import html
import sys
from pathlib import Path

import streamlit as st


# ==========================================================
# PROJECT PATH
# ==========================================================
```

```
PROJECT_ROOT = Path(__file__).resolve().parent.parent

if str(PROJECT_ROOT) not in sys.path:
    sys.path.insert(0, str(PROJECT_ROOT))


# ============================================================
# PROJECT IMPORTS
# ============================================================

from auth.auth import authenticate_user, register_user
from dashboard.components.styles import APP_CSS
from dashboard.utils.data_loader import load_all_data
from dashboard.views import (
    render_about,
    render_comparison,
    render_country_analysis,
    render_data_explorer,
    render_global_analysis,
    render_india_analysis,
    render_insights_page,
    render_overview,
    render_peak_analysis,
    render_rankings,
    render_trends,
    render_vaccination,
)
from database.database import init_db, log_activity


# ============================================================
# HTML HELPERS
# ============================================================

def render_html(content: str) -> None:
    """Render multiline HTML on the main page with proper dedenting."""
    # Remove ALL leading whitespace from each line to prevent Markdown code block
interpretation
    lines = [line.lstrip() for line in content.split('\n')]
    cleaned = '\n'.join(lines).strip()
    st.markdown(cleaned, unsafe_allow_html=True)


def render_sidebar_html(content: str) -> None:
    """Render multiline HTML in the sidebar with proper dedenting."""
    # Remove ALL leading whitespace from each line to prevent Markdown code block
interpretation
    lines = [line.lstrip() for line in content.split('\n')]
    cleaned = '\n'.join(lines).strip()
    st.sidebar.markdown(cleaned, unsafe_allow_html=True)
```

```python
# =========================================================
# STREAMLIT CONFIGURATION
# =========================================================

st.set_page_config(
    page_title="COVID-19 Pandemic Trend Analysis",
    page_icon="🧬",
    layout="wide",
    initial_sidebar_state="expanded",
)


# =========================================================
# APPLICATION CSS
# =========================================================

st.markdown(
    APP_CSS,
    unsafe_allow_html=True,
)


# =========================================================
# DATABASE
# =========================================================

init_db()


# =========================================================
# DASHBOARD PAGES
# =========================================================

PAGES = [
    "Overview",
    "Global Analysis",
    "Country Analysis",
    "India Analysis",
    "Pandemic Trends",
    "Peak Analysis",
    "Vaccination Analysis",
    "Country Comparison",
    "Country Rankings",
    "Data Explorer",
    "Insights",
    "About",
]
```

```python
# ==========================================================
# SESSION INITIALIZATION
# ==========================================================

def _init_session() -> None:
    """Initialize application session state."""

    if "authenticated" not in st.session_state:
        st.session_state.authenticated = False

    if "user" not in st.session_state:
        st.session_state.user = None

    if "auth_mode_selector" not in st.session_state:
        st.session_state.auth_mode_selector = "Sign in"

    if "dashboard_navigation" not in st.session_state:
        st.session_state.dashboard_navigation = "Overview"


# ==========================================================
# AUTHENTICATION LANDING PAGE
# ==========================================================

def render_auth() -> None:
    """Render the polished medical-tech authentication page."""

    # ======================================================
    # PAGE INTRO / BRAND
    # ======================================================

    render_html(
        """
        <div class="auth-page-shell">

            <div class="auth-brand-row">

                <div class="auth-brand-icon">
                    <span>+</span>
                </div>

                <div>
                    <div class="auth-brand-name">
                        PANDEMIC ANALYTICS
                    </div>

                    <div class="auth-brand-subtitle">
                        Clinical surveillance intelligence
                    </div>
                </div>
```

```
        </div>

    </div>
    """
)

# ======================================================
# MAIN AUTHENTICATION LAYOUT
# ======================================================

left_col, right_col = st.columns(
    [1.15, 0.85],
    gap="large",
)

# ======================================================
# LEFT MEDICAL-TECH PANEL
# ======================================================

with left_col:

    render_html(
        """
        <div class="medical-hero">

            <div class="hero-grid-pattern"></div>

            <div class="hero-content">

                <div class="hero-status">
                    <span class="live-dot"></span>
                    PANDEMIC SURVEILLANCE PLATFORM
                </div>

                <h1 class="medical-title">
                    COVID-19
                    <br>
                    <span>Pandemic Trend Analysis</span>
                </h1>

                <p class="medical-copy">
                    Explore global pandemic trends, country-level
                    surveillance, vaccination coverage, peak intensity,
                    comparative analysis and data-driven insights.
                </p>

                <div class="medical-feature-list">

                    <div class="medical-feature">
                        <div class="feature-icon">↗</div>
```

```
                <div>
                    <strong>Global surveillance</strong>
                    <span>
                        Monitor reported pandemic trends
                    </span>
                </div>
            </div>

            <div class="medical-feature">
                <div class="feature-icon">◎</div>

                <div>
                    <strong>Country intelligence</strong>
                    <span>
                        Compare locations and outcomes
                    </span>
                </div>
            </div>

            <div class="medical-feature">
                <div class="feature-icon">ϟ</div>

                <div>
                    <strong>Data-driven insights</strong>
                    <span>
                        Identify peaks and emerging patterns
                    </span>
                </div>
            </div>

        </div>

    </div>

    <!-- MEDICAL DATA VISUAL -->

    <div class="medical-visual">

        <div class="visual-ring ring-one"></div>
        <div class="visual-ring ring-two"></div>
        <div class="visual-ring ring-three"></div>

        <div class="visual-core">

            <div class="core-cross">
                +
            </div>

            <div class="core-label">
                COVID
            </div>
```

```
        <div class="core-small">
            GLOBAL DATA
        </div>

    </div>

    <div class="data-node node-one">
        <span class="node-dot"></span>
        CASES
    </div>

    <div class="data-node node-two">
        <span class="node-dot"></span>
        TRENDS
    </div>

    <div class="data-node node-three">
        <span class="node-dot"></span>
        VACCINATION
    </div>

    <div class="floating-panel panel-top">

        <span class="panel-label">
            GLOBAL STATUS
        </span>

        <strong>
            SURVEILLANCE
        </strong>

        <small>
            Data monitoring active
        </small>

    </div>

    <div class="floating-panel panel-bottom">

        <div class="mini-chart">

            <span style="height: 24%"></span>
            <span style="height: 38%"></span>
            <span style="height: 31%"></span>
            <span style="height: 55%"></span>
            <span style="height: 44%"></span>
            <span style="height: 70%"></span>
            <span style="height: 82%"></span>

        </div>
```

```
                <div class="panel-bottom-text">

                    <strong>
                        Trend intelligence
                    </strong>

                    <small>
                        Multi-country analysis
                    </small>

                </div>

            </div>

        </div>

        <div class="hero-footer-line">

            <span>GLOBAL</span>
            <span>COUNTRY</span>
            <span>VACCINATION</span>
            <span>PEAK ANALYSIS</span>

        </div>

    </div>
    """
)

# ======================================================
# RIGHT AUTHENTICATION PANEL
# ======================================================

with right_col:

    render_html(
        """
        <div class="auth-card">

            <div class="auth-account-label">
                ACCOUNT ACCESS
            </div>

            <h2 class="auth-card-title">
                Welcome back
            </h2>

            <p class="auth-card-subtitle">
                Sign in to continue to your analytics workspace.
            </p>
```

```
    </div>
    """
)

# ==================================================
# AUTH MODE SELECTOR
# ==================================================

mode = st.radio(
    "Select an option",
    ["Sign in", "Create account"],
    horizontal=True,
    key="auth_mode_selector",
)

# ==================================================
# SIGN IN
# ==================================================

if mode == "Sign in":

    render_html(
        """
        <div class="auth-section-heading">

            <div class="auth-section-title">
                Sign in
            </div>

            <div class="auth-section-description">
                Enter your credentials to access the dashboard.
            </div>

        </div>
        """
    )

    with st.form(
        "login_form",
        clear_on_submit=False,
    ):

        username = st.text_input(
            "Username",
            placeholder="Enter your username",
        )

        password = st.text_input(
            "Password",
            type="password",
```

```
                placeholder="Enter your password",
            )

            submitted = st.form_submit_button(
                "Sign in to dashboard",
                width="stretch",
            )

        if submitted:

            username = username.strip()

            ok, message, profile = authenticate_user(
                username,
                password,
            )

            if ok:

                st.session_state.authenticated = True
                st.session_state.user = profile

                if profile and profile.get("id") is not None:

                    try:
                        log_activity(
                            profile["id"],
                            "login",
                        )
                    except Exception:
                        pass

                st.rerun()

            else:

                st.error(message)

    # ==================================================
    # CREATE ACCOUNT
    # ==================================================

    else:

        render_html(
            """
            <div class="auth-section-heading">

                <div class="auth-section-title">
                    Create account
                </div>
```

```
            <div class="auth-section-description">
                Set up your account to access the analytics workspace.
            </div>

        </div>
        """
    )

    with st.form(
        "register_form",
        clear_on_submit=False,
    ):

        full_name = st.text_input(
            "Full name",
            placeholder="Enter your full name",
        )

        username = st.text_input(
            "Username",
            placeholder="Choose a username",
        )

        email = st.text_input(
            "Email",
            placeholder="Enter your email address",
        )

        password = st.text_input(
            "Password",
            type="password",
            placeholder="Minimum 8 characters",
        )

        confirm = st.text_input(
            "Confirm password",
            type="password",
            placeholder="Re-enter your password",
        )

        submitted = st.form_submit_button(
            "Create account",
            width="stretch",
        )

    if submitted:

        ok, message = register_user(
            full_name.strip(),
            username.strip(),
```

```
            email.strip(),
            password,
            confirm,
        )

        if ok:

            st.success(message)

            st.session_state.auth_mode_selector = "Sign in"

            st.rerun()

        else:

            st.error(message)

# ==================================================
# SECURITY NOTE
# ==================================================

render_html(
    """
    <div class="auth-security">

        <div class="security-icon">
            🔒
        </div>

        <div>

            <strong>
                Secure workspace
            </strong>

            <span>
                Your credentials are protected using password hashing.
            </span>

        </div>

    </div>
    """
)

# ==================================================
# AUTH FOOTER
# ==================================================

render_html(
    """
```

```
            <div class="auth-footer">

                <span>
                    COVID-19 Pandemic Trend Analysis
                </span>

                <span class="footer-divider">
                    •
                </span>

                <span>
                    Academic analytics platform
                </span>

            </div>
            """
        )


# ===========================================================
# SIDEBAR
# ===========================================================

def render_sidebar() -> str:
    """Render the dashboard sidebar."""

    user = st.session_state.user or {}

    full_name = html.escape(
        str(user.get("full_name", "User"))
    )

    username = html.escape(
        str(user.get("username", "user"))
    )

    # =====================================================
    # BRAND
    # =====================================================

    render_sidebar_html(
        """
        <div class="sidebar-brand">

            <div class="sidebar-brand-icon">
                +
            </div>

            <div>

                <div class="sidebar-brand-kicker">
```

```
                SURVEILLANCE
            </div>

            <div class="sidebar-brand-title">
                COVID-19
            </div>

            <div class="sidebar-brand-subtitle">
                Pandemic Trend Analysis
            </div>

        </div>

    </div>
    """
)

# ========================================================
# USER PROFILE
# ========================================================

initial = (
    full_name.strip()[0].upper()
    if full_name.strip()
    else "U"
)

render_sidebar_html(
    f"""
    <div class="sidebar-user-card">

        <div class="user-avatar">
            {html.escape(initial)}
        </div>

        <div class="user-details">

            <strong>
                {full_name}
            </strong>

            <span>
                @{username}
            </span>

        </div>

    </div>
    """
)
```

```python
# =========================================================
# NAVIGATION LABEL
# =========================================================

render_sidebar_html(
    """
    <div class="sidebar-section-label">
        NAVIGATION
    </div>
    """
)

# =========================================================
# NAVIGATION
# =========================================================

page = st.sidebar.radio(
    "Navigation",
    PAGES,
    key="dashboard_navigation",
    label_visibility="collapsed",
)



# =========================================================
# SIGN OUT
# =========================================================

if st.sidebar.button(
    "Sign out",
    width="stretch",
    key="sign_out_button",
):

    if user.get("id") is not None:

        try:
            log_activity(
                user["id"],
                "logout",
            )
        except Exception:
            pass

    st.session_state.authenticated = False
    st.session_state.user = None
    st.session_state.auth_mode_selector = "Sign in"

    st.rerun()
```

```
    # =====================================================
    # SIDEBAR STATUS
    # =====================================================

    render_sidebar_html(
        """
        <div class="sidebar-status">
            <span class="status-dot"></span>
            Protected workspace - SQLite session
        </div>
        """
    )

    return page


# =========================================================
# DASHBOARD CONTENT
# =========================================================

def render_dashboard(page: str) -> None:
    """Load datasets and render the selected dashboard page."""

    try:

        data = load_all_data()

    except Exception as exc:

        st.error(
            "The dashboard could not load the required datasets."
        )

        st.exception(exc)

        st.stop()

    # =====================================================
    # DATASETS
    # =====================================================

    country_df = data["country"]
    global_df = data["global"]
    peaks_df = data["peaks"]
    vaccination_df = data["vaccination"]

    # =====================================================
    # PAGE ROUTING
    # =====================================================

    if page == "Overview":
```

```python
    render_overview(
        country_df,
        global_df,
        peaks_df,
    )

elif page == "Global Analysis":

    render_global_analysis(
        global_df,
    )

elif page == "Country Analysis":

    render_country_analysis(
        country_df,
        vaccination_df,
    )

elif page == "India Analysis":

    render_india_analysis(
        country_df,
        global_df,
        peaks_df,
        vaccination_df,
    )

elif page == "Pandemic Trends":

    render_trends(
        global_df,
    )

elif page == "Peak Analysis":

    render_peak_analysis(
        peaks_df,
        country_df,
    )

elif page == "Vaccination Analysis":

    render_vaccination(
        country_df,
        vaccination_df,
    )

elif page == "Country Comparison":
```

```
    render_comparison(
        country_df,
    )

elif page == "Country Rankings":

    render_rankings(
        country_df,
    )

elif page == "Data Explorer":

    render_data_explorer(
        country_df,
        global_df,
    )

elif page == "Insights":

    render_insights_page(
        country_df,
        global_df,
        peaks_df,
    )

elif page == "About":

    render_about(
        country_df,
        global_df,
    )

# =====================================================
# DASHBOARD FOOTER
# =====================================================

render_html(
    """
    <div class="dashboard-footer">

        <span>
            COVID-19 Pandemic Trend Analysis
        </span>

        <span class="footer-divider">
            •
        </span>

        <span>
            Reported surveillance data from project CSV files
        </span>
```

```
        </div>
        """
    )


# ============================================================
# MAIN
# ============================================================

def main() -> None:
    """Application entry point."""

    _init_session()

    # ======================================================
    # AUTHENTICATION PAGE
    # ======================================================

    if not st.session_state.authenticated:

        render_auth()

        return

    # ======================================================
    # DASHBOARD
    # ======================================================

    page = render_sidebar()

    render_dashboard(page)


# ============================================================
# RUN APPLICATION
# ============================================================

if __name__ == "__main__":
    main()
```